

Harshith Kurakula

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SUMMARY

Software Engineer with 3+ years architecting and operating distributed systems in production cloud environments, with ownership spanning database infrastructure, reliability, and incident response. Strong foundation in data structures, algorithms, and software design, with recent work in AI-integrated multi-agent systems. Comfortable moving fluidly across the stack, with a bias toward shipping.

EDUCATION

Arizona State University

Tempe, AZ, USA

Master of Science in Computer Science, **Current GPA - 4.00 / 4.00**

August 2025 – December 2026

Relevant Coursework: Data Mining, Statistical Machine Learning, Data Visualization, Semantic Web Mining, Knowledge Representation

Vellore Institute of Technology

Vellore, Tamil Nadu, India

Bachelor of Technology in Computer Science and Engineering, **GPA - 3.84 / 4.00**

July 2018 – June 2022

Relevant Coursework: Data Structures, Algorithms, Database Systems, Object-Oriented Programming, Web Development

SKILLS

Programming Languages:	Python, Java, SQL, C/C++, JavaScript, TypeScript
Cloud, DevOps & Databases:	AWS, GCP, Docker, Kubernetes, Terraform, CI/CD, Datadog, Splunk, MySQL, PostgreSQL
Web Development:	HTML, CSS, Django, FastAPI, REST APIs, GraphQL, Node.js, React, Next.js
ML & Data Tools:	NumPy, Pandas, Scikit-learn, TensorFlow/Keras, PyTorch, OpenCV, Tableau

EXPERIENCE

Chargebee Technologies

Chennai, Tamil Nadu, India

Senior Software Engineer

April 2025 – July 2025

- Architected a unified AWS RDS connection pooling system, consolidating schema-level pools into a single shared pool per instance, reducing **idle connection overhead by 20%** and improving throughput, utilization, and stability across production microservices.
- Drove a centralized database configuration management platform enforcing consistent configuration propagation across all microservices, reducing **manual provisioning effort by 40%** and cutting environment-drift incidents during infrastructure onboarding.

Software Engineer

August 2022 – March 2025

- Led AWS RDS Blue-Green deployment upgrade initiatives across critical production databases, coordinating planning, execution, and validation to achieve near-zero downtime, **reduce service disruption by 90%**, and preserve complete data integrity across systems.
- Engineered an automated schema comparison tool in Java and SQL to detect database inconsistencies across environments, capturing all schema mismatches while **reducing manual debugging time by 30%** and improving overall production reliability.
- Launched an internal API feature for a database migration tool to automate schema changes following RDS provisioning and sharding activities, **reducing manual intervention by 25%** and improving system efficiency and reliability.
- Scaled SRE best practices across cross-functional engineering, platform, and product teams by institutionalizing incident post-mortems and runbooks, **reducing incident recurrence by 30%** and enabling reliable, scalable multi-region deployments.

Software Engineer Intern

January 2022 – July 2022

- Engineered a centralized observability and monitoring dashboard in Python integrating Splunk, ELK Stack, and database logs, enabling proactive incident detection and **reducing Mean Time to Resolve (MTTR) by 25%** while **maintaining 99.9% system availability**.
- Automated infrastructure provisioning across AWS RDS, DynamoDB, and CloudFormation, **reducing manual configuration effort by 40%**, **accelerating deployment time by 30%**, and improving cross-environment reliability and consistency.

PROJECTS

Multi-Agent Travel Itinerary Planner (Python, LangGraph, Streamlit, Vertex AI)

- Architected an 8-component multi-agent LLM system using LangGraph, coordinating specialized agents to generate constraint-aware, multi-day travel itineraries, raising final pass rate from **4.4% to 31%** over a single-agent GPT-4 Turbo baseline on identical data.
- Built a two-stage repair pipeline (mechanical pre-repair + up to 3 LLM-driven rounds) with agent-attribution logging for targeted routing, achieving **100% delivery** and **94% commonsense constraint satisfaction** across 180 benchmark queries at 4.3 LLM calls/query.

Multimodal Image Retrieval System (Python, Keras, FastAPI, React)

- Designed and implemented a multimodal deep learning retrieval pipeline leveraging ResNet (CNN) for visual feature extraction and LSTM-based text embeddings, **achieving 80–85% retrieval accuracy** by effectively aligning visual and semantic representations.
- Built an end-to-end image–text retrieval workflow with a FastAPI backend and React frontend, integrating advanced text preprocessing and image feature analysis to **significantly outperform traditional baseline models** in search relevance and precision.